Hoang Dang

Boston, MA

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Experience

MIT Computer Science and AI Lab - Kellis Lab — **Boston, MA**
Research Assistant (Advisor: Manolis Kellis) — *Dec 2025 – Present*

- Lead the integration of Mantis4Hospital Vertical, and developed an ontology-based knowledge graph to mapped all EHR codes and insurance claims, increase revenue by 20%
- Built MantisFlybase platform for single cells RNA-sequencing for the intergration of FlyBase data.
- Design and implement security dashboards and performance frameworks to dynamically detect security issues across the platform, reducing security cost by 55%

Spaulding Neuromodulation Center/ Harvard Medical School — **Boston, MA**
Research Scholar (Advisor: Kevin Pachecco-Barrios) — *March 2026 – Present*

- Built **Valor** and **Integral**, mobile platforms for continuous monitoring and longitudinal research of outcomes in 40+ concussion and burn-injury patients across 3 clinical sites, and **Ecological Momentary Assessment** tooling that captured 40 patient-reported data points

Lockheed Martin — **Remote**
Software Engineer (Contract) — *August 2024 – May 2025*

- Developed a computer vision system in Python and PyTorch to detect cognitive load using emotions detection and body language, achieving accuracy of 85%

E3 Company — **Remote**
Software Engineer (Contract) — *Jan 2025 – May 2025*

- Designed and developed RESTful APIs and backend services to ingest, store, and serve monitoring data with low latency in JavaScript.
- Built dashboards and alerting features that enabled operators to monitor equipment health and respond to critical events in JavaScript and React.

Projects

BurnX — **1st Place, MIT GrandHack 2026**
Burn Monitoring Device with AI Risk Detection — *2026*

- Developed an iOS mobile application in Swift, integrating the computer vision model and Twilio SMS notifications to alert nursing staff of potential high-risk cases.

FoldForward — **1st Place, HackNation AI**
AI Co-Scientist for Protein Structure Prediction — *2026*

- Built an agentic AI co-scientist that generates, runs, and ranks structure-prediction experiments for the $\alpha 7$ nAChR receptor.

Skills

Languages: C#, HTML/CSS, Java, JavaScript, LaTeX, Python, SQL
Software: Atlassian (Bitbucket, Jira, Confluence), AWS (DynamoDB, EC2, Lambda, RDS, Redshift, S3, SQS), Microsoft (Azure DevOps, Visual Studio), DigitalOcean, Django, Heroku, IntelliJ IDEA, Selenium

Education

Gannon University — **Erie, PA**
B.S. Computer Science (GPA: 3.96/4.00) — *August 2021 – May 2025*